

Maximum Sequence Length MASH Digital Delta-Sigma Modulators

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Abstract—This paper presents a modified structure for the first-order digital delta-sigma modulator (DDSM) which yields the maximum sequence length (N) for all constant digital inputs and for all initial conditions. Using this structure, Multistage noise-Shaping (MASH) DDSMs are developed which, for a constant input, always have a sequence length of N^l , regardless of the initial conditions and input value, where N is the sequence length of each stage and l is the number of the first-order stages used in the MASH. The performance is confirmed mathematically, by simulation, and experimentally.

Index Terms—Digital delta-sigma modulator (DDSM), digital-to-analog converters (DACs), fractional- N frequency synthesizer, maximum sequence length, multistage noise shaping (MASH).

I. INTRODUCTION

THE DIGITAL delta-sigma modulator (DDSM) shown in Fig. 1 is a discrete-time deterministic dynamical system with a digital input and a digital output. The DDSM receives an n_0 -bit digital input sequence x and delivers an m -bit digital output sequence y , as shown. Generally, m is significantly less than n_0 . In a practical situation such as the front-end of an over-sampling digital-to-analog converter (DAC) in a CD player, n_0 and m might be 14 and 1, respectively.

Fig. 2 illustrates the operation of the DDSM in the frequency domain. Assume that we have applied to the input of the modulator a high resolution low-frequency tone f_0 encoded over n_0 -bits with the spectrum given in Fig. 2(a). As shown in Fig. 2(b), the DDSM delivers a low resolution sequence to the output whose spectrum contains the original spectrum of the input signal. Note that the spectrum of the low resolution output sequence contains the full information of the high resolution input sequence. Ideally, the resulting quantization error is whitened and filtered by the DDSM so that its power is moved away from the signal through a process called noise shaping. The shaped quantization error (noise) appears in the output spectrum as shown in Fig. 2(b). Note that the power of the quantization noise is concentrated away from f_0 and around $f_s/2^l$, thereby maximizing the signal-to-noise ratio close to dc.

If the output signal of the DDSM is applied to an ideal continuous-time low-pass filter [Fig. 2(c)] that passes only the low-frequency content of the modulator output, then the original signal can be recovered from the low resolution output signal with a



Fig. 1. Block diagram of the DDSM with n_0 -bit input $x[n]$ and m -bit output $y[n]$.

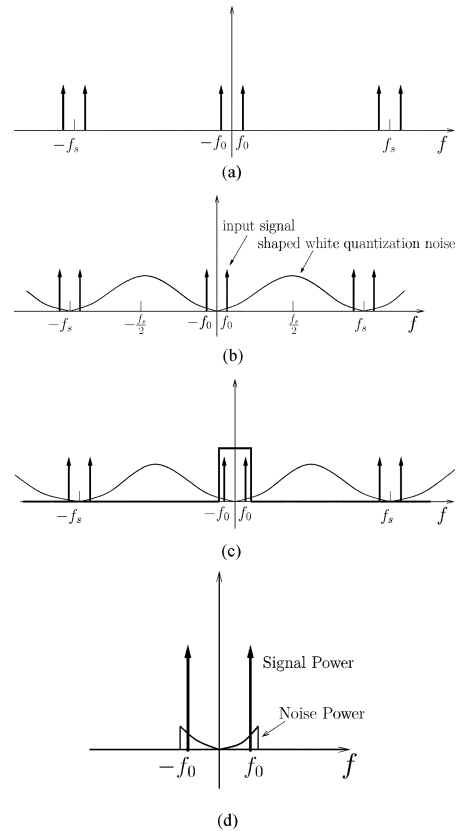


Fig. 2. Spectra in a DDSM with a sinusoidal input. (a) Power spectrum of the input tone. (b) Power spectrum of the output sequence. (c) The low-pass filter is applied to the DDSM output. (d) The low-pass filter output spectrum (enlarged).

high signal-to-quantization noise ratio. Fig. 2(d) shows the spectrum at the output of the low-pass filter (enlarged). Note that the signal power of the filtered output is much higher than the noise power content.

DDSMs may be implemented in a number of different ways [1]. An important application for DDSMs is in fractional- N frequency synthesizers [2]–[9]. For unmodulated synthesizers, the input to the DDSM is a constant digital word. In this paper, we investigate constant input Multistage noise SHaping (MASH) DDSMs incorporating first-order error feedback modulators (EFMs).

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¹ f_s is the clock frequency of the DDSM.

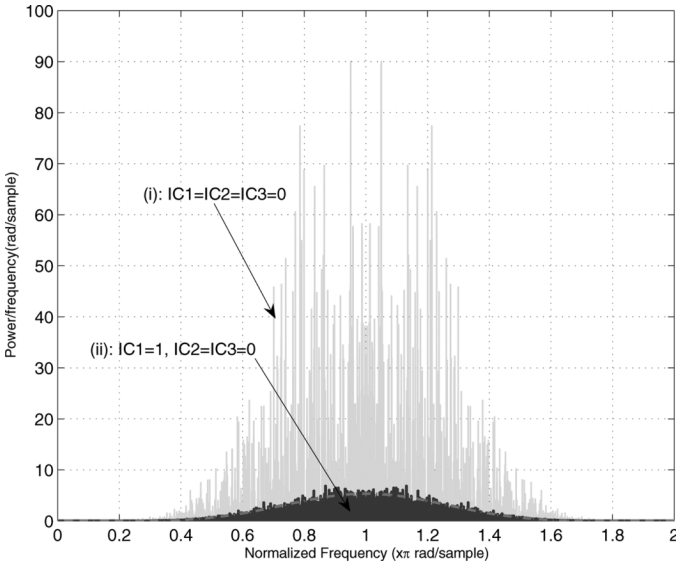


Fig. 3. Effect of short nonwhite sequences in a 14-bit MASH1-1-1 DDSM; the axes are shown with linear scales. IC_i is the initial condition on stage i . (i) $IC_1 = IC_2 = IC_3 = 0$; sequence length is 4096. (ii) $IC_1 = 1, IC_2 = IC_3 = 0$; sequence length is 32768. The dashed curve shows the shaped white quantization noise.

Since the DDSM is a finite-state machine (FSM), it always produces a periodic output sequence (a cycle) when the input is a constant. In particular, the quantization noise sequence is periodic. The period of the noise sequence depends on the input, the initial conditions, and the architecture of the DDSM. For some inputs and for some initial conditions, the sequences may become relatively short. When the period is short, the power of the quantization noise is distributed among a small number of tones and the spectrum deviates significantly from the smooth curve shown in Fig. 2(b).

To illustrate this problem, we refer to Fig. 3 which shows the results of MATLAB simulations. A MASH DDSM with $n_0 = 14$ and $m = 3$ is simulated with a constant input but with two different sets of initial conditions. The MASH1-1-1 structure we consider consists of three first-order EFMs whose internal registers can be initialized (seeded) to any digital value in the range $\{0, 1, 2, \dots, (2^{n_0} - 1)\}$. The input to the modulator in both cases is a constant $X = 8$. The plots are shown with linear axes and the normalized frequency 1 corresponds to $f_s/2$. The dc component has been removed from each of the plots.

Plot (i) shows the simulation result when the seeds of all stages are set to zero. Plot (ii) shows the power spectrum of the output sequence when the initial state (seed) of the first stage is set to 1 and the seeds of the other stages are set to zero. Note that the spectrum of plot (ii) can be approximated by a smooth curve, the dashed curve, representing the shaped white noise pushing the quantization noise power gently toward the high frequencies. By contrast, the spectrum shown in plot (i) contains some high power tones resulting from insufficient randomization of the quantization noise sequence.

This example illustrates the problems associated with short nonwhite sequences in a DDSM. Two classes of techniques have been developed to whiten the quantization noise: stochastic

TABLE I
SEQUENCE LENGTHS IN CONVENTIONAL MASH [12] AND [13]

Modulator order, l	Minimum Sequence Length with odd initial condition on the first stage	Maximum Sequence Length with odd initial condition on the first stage
2	$2^{(n_0-1)}$	$2^{(n_0+1)}$
3	$2^{(n_0+1)}$	$2^{(n_0+1)}$
4	$2^{(n_0+1)}$	$2^{(n_0+2)}$
5	$2^{(n_0+2)}$	$2^{(n_0+2)}$

and deterministic. The “stochastic” approach to maximizing sequence lengths is to use a “random”² dither sequence to disrupt periodic cycles [1], [10]. Dithering breaks up the cycles and increases the effective sequence length, resulting in smooth noise-shaped spectra. While the “stochastic” solution increases the sequence length, as required, it inherently adds noise to the spectrum; care must be taken to minimize the contribution of this additional noise. A state of the art solution is to add shaped pseudorandom LSB dither to the input of the MASH DDSM [10].

By contrast, the deterministic approach to whitening the quantization noise is to guarantee maximum sequence lengths by design. One approach is to avoid known short sequences by setting the initial conditions of the internal registers of the stages. Recent work [11]–[13] has addressed this technique. In [11], [12], and [13], seeding the first stage to an odd value in the MASH DDSM has been shown, by empirical observations and mathematically, to maximize the sequence length. Another way of maximizing the sequence length is to use a prime modulus quantizer [14].

In this paper, we consider the deterministic approach and present a novel strategy for maximizing sequence lengths in MASH DDSMs by modifying the constituent first-order EFMs.

State-of-the-art quantitative estimates of the sequence lengths for the ditherless deterministic MASH DDSM are summarized in Table I [12]. The first column shows the order of the MASH DDSM. The initial condition of the first stage of the MASH is assumed to be set to an odd value. The values reported in the second column are the guaranteed minimum sequence lengths. The third column shows the maximum achievable sequence lengths. According to Table I, the sequence length is of the order of 2^{n_0} , provided that the initial condition of the first stage is set to an odd value [12], [13]. In the case of the third-order modulator, all inputs yield maximum length sequences.

We will show that our modified MASH structure guarantees a sequence length of order 2^{ln_0} (rather than 2^{n_0}) that is achieved for *all* dc inputs³ and for *all* initial conditions. In particular, the maximum sequence length for our third-order DDSM is approximately 2^{3n_0} , compared to 2^{n_0+1} in the conventional case [12].

The paper is organized as follows. In Section II, the MASH DDSM, is reviewed. In Section III, we introduce a modified first-order DDSM in order to achieve the longest possible sequences for all dc inputs and for all initial conditions. Based on this structure, we introduce a modified MASH structure in Section IV which guarantees a sequence length of N^l , where N is the maximum sequence length of the first stage and l is the number of first-order stages introduced in Section III. In

²The “random” sequence might be generated by a pseudorandom or chaotic source.

³input $x[n] = X \in \{1, \dots, (M - a - 1)\}$.

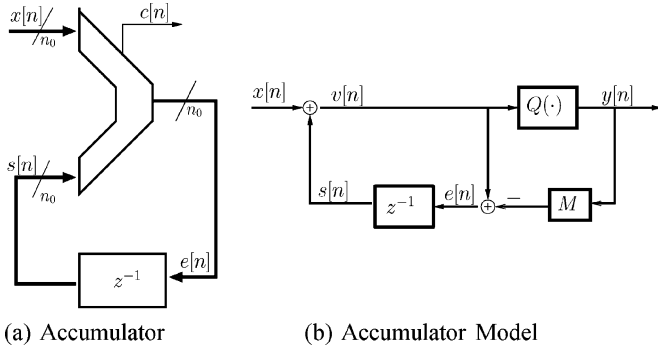


Fig. 4. Digital accumulator used in MASH DDSM (a) and its model (b).

Section V, we present simulation and experimental results for a state of the art modulator and the modified modulator and we compare their performances. Finally, we prove the maximum sequence length mathematically in Appendices I and II.

II. STATE-OF-THE-ART MASH STRUCTURE

Before exploring our new structure, we first review a state of the art MASH DDSM. The basic building block of the MASH DDSM we consider is a digital accumulator. The accumulator and its model are shown in Fig. 4. Referring to Fig. 4(a), the input to the accumulator $x[n]$ is a digital word with n_0 bits. In the case of an unmodulated $\Delta\Sigma$ fractional- N frequency synthesizer, for example, the input is a constant X which sets the desired fraction, i.e. $X/2^{n_0}$ [2], [3]. The register, labeled z^{-1} , delays the error sequence $e[n]$ by one clock cycle. The delayed sequence $s[n]$ is added to the input sequence $x[n]$. When the accumulator overflows, it generates a carry bit $c[n]$. The time average of the carry-out sequence, consisting of digital 0 s and 1 s, is equal to the time average of the input sequence. All the signals shown in Fig. 4 have integer values.

Fig. 4(b) is a model of the digital accumulator shown in Fig. 4(a). The output $y[n]$ corresponds to the carry-out signal $c[n]$. The quantizer block $Q(\cdot)$ corresponds to the overflow operation of the digital accumulator. If the signal $v[n]$, which is the sum of the two digital words $x[n]$ and $s[n]$, is greater than or equal to $M (= 2^{n_0})$, overflow occurs and $y[n]$ will be 1; otherwise it is 0. We write mathematically

$$y[n] = Q(v[n]) = \begin{cases} 0, & v[n] < M \\ 1, & v[n] \geq M \end{cases} \quad (1)$$

where $M = 2^{n_0}$. The difference between $v[n]$ and $My[n]$ is fed back to the input via a register (labeled z^{-1}). If $y[n]$ is 0 then $e[n] = v[n]$; otherwise $e[n] = v[n] - M$. Therefore, we have

$$e[n] = v[n] \bmod M. \quad (2)$$

The gain block M is shown to illustrate the weight of $y[n]$. This structure is known as an EFM because the error signal $s[n]$ is fed back to the input of the accumulator [12]. In the following, we observe that the modulator shapes the spectrum of the quantizer error (noise), thereby distributing the power into higher frequency components.

To simplify the analysis, one can build an approximate linear model of the modulator in which the quantizer is replaced by

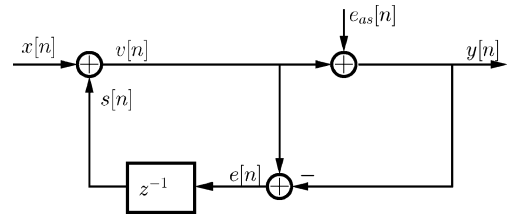


Fig. 5. Block diagram of a first-order digital EFM when the quantizer is replaced by an additive noise source $e_{as}[n]$. All the signals are normalized to 1 with respect to $M = 2^{n_0}$.

an additive quantization noise source $e_{as}[n]$, as shown in Fig. 5. We can write the linear z -domain output $Y(z)$ in terms of $X(z)$ and $E_{as}(z)$ as follows:

$$Y(z) = X(z) + (1 - z^{-1})E_{as}(z) \quad (3)$$

where

$$E_{as}(z) = -E(z). \quad (4)$$

From (3), we conclude that the output contains two components: the first term is the input sequence and the second term, which is the output quantization error, is the quantizer error $E_{as}(z)$, shaped by the filter $(1 - z^{-1})$. If we assume that the quantizer error sequence $e_{as}[n]$ has a white noise spectrum (equal power in each tone), then the spectrum of the output quantization error [the second term in (3)] is a smooth curve shaped by the filter $(1 - z^{-1})$. In this case, the smooth noise-shaping approximation illustrated in Fig. 2(b) might be realized.

When the input is a constant, the above assumption is not always correct. Since the modulator is a deterministic finite state machine, the quantization error sequence is periodic when the input is constant. We have shown in [13] that the period of this first-order structure depends explicitly on the value of the input. For some inputs, the sequence length is relatively short; in this case, the output quantization error spectrum is not a smoothly shaped curve. Instead, it contains a small number of tones deviating significantly from the smooth noise-shaping curve.

In order to achieve higher order noise shaping [1], higher order MASH modulators have been developed. These incorporate a cascade of first-order EFMs (EFM1 block), as shown in Fig. 6. In MASH, the error of each stage ($e_i[n]$) is fed to the next stage and the carry-out of each stage ($y_i[n]$) is fed to the noise cancellation network to eliminate the intermediate quantization noises. $s_i[0]$ denotes the initial condition on accumulator i . The spectrum of the final output $y[n]$ includes the spectrum of the input signal $x[n]$ in addition to the shaped quantization noise power of the last stage.

If we replace the quantizer of each first-order stage with an additive noise source $e_{asi}[n]$ like that described in the case of the first-order modulator shown in Fig. 3, the linear input-output relationship in the z -domain will be [2]

$$Y(z) = X(z) + (1 - z^{-1})^L E_{asl}(z) \quad (5)$$

where E_{asl} is the quantization error of the last stage. From (5), we find that the spectrum of the input signal appears directly in

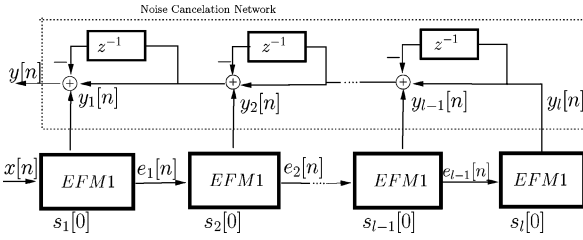


Fig. 6. Block diagram of the digital MASH DSM. EFM1 Blocks are the structure shown in Fig. 4(b).

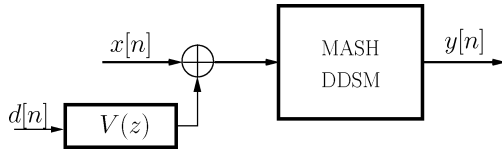


Fig. 7. Block diagram of the simulated dithered-MASH DDSM.

the output spectrum. The second term is the quantization noise of the last stage shaped by the filter $(1 - z^{-1})^l$, where l is the number of first-order stages.

Although the digital MASH configuration performs higher order noise shaping, it still suffers from the problem of the non-white short quantization noise sequences. Despite its more complex structure, the sequence length can still be very short for many values of inputs and initial conditions (see Fig. 3).

As mentioned in the previous section, “stochastic” and “deterministic” approaches can be used to maximize sequence lengths. Here we provide two examples to illustrate the relative merits of these methods.

A. Stochastic Approach: Dithering

To illustrate dithering, we consider two strategies described in [10]. A 1-bit pseudorandom dither sequence $d[n]$ is added to the LSB of the dc input $x[n]$ after being applied to a filter $V(z)$ (see Fig. 7). Fig. 8 shows the spectra of the resulting output sequences for a 9-bit MASH1–1–1 DDSM with $X = 1$ for three cases: i) without dither, $V(z) = 0$; ii) LSB dither, $V(z) = 1$; and iii) first-order shaped LSB dither, $V(z) = (1 - z^{-1})$. The plots are shown on log scales. $f_s/2$ has been normalized to 1 on the x -axis.

From the figure, we see that the ditherless modulator exhibits strong tonal behavior due to the relatively short periodic output sequence ($L_s = 2^{10}$). With a pseudorandom LSB dither sequence of length $2^{21} - 1$ applied, the spectrum is smoother, as expected, but the noise floor rises. The low-frequency performance can be improved by first-order shaping the dither. Increasing the order of the filter $V(z)$ may improve the low-frequency performance but a spur free noise-shaping spectrum cannot be guaranteed [10].

B. Deterministic Approach: Maximum Sequence Length by Design

Recent work has addressed the problem of guaranteeing an acceptable minimum sequence length for all constant inputs [12] without dither, the objective being to whiten the spectra without compromising the low-frequency performance. The authors of [12] have considered DDSMs as finite state machines

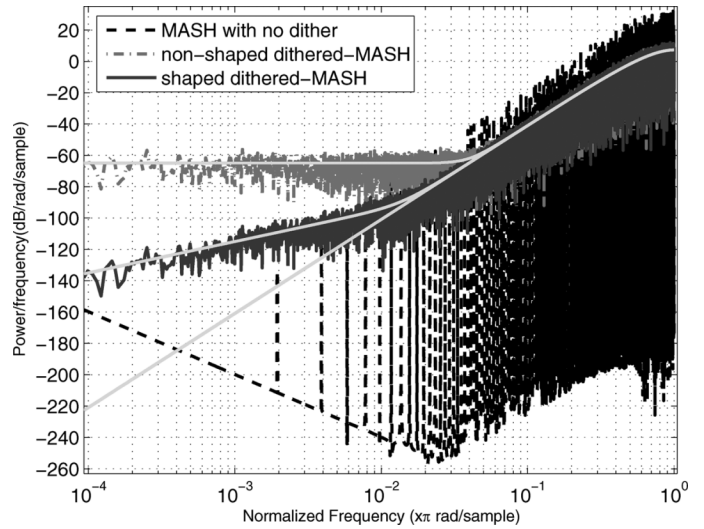


Fig. 8. Output spectra of a 9-bit MASH1–1–1 DDSM for three cases with constant input $X = 1$: (i) MASH without dithering, (ii) MASH with LSB dither added to the input and (iii) MASH with first-order shaped LSB dither added to the input. The solid curves represent the shaped white quantization noise, including the dither contributions where applied.

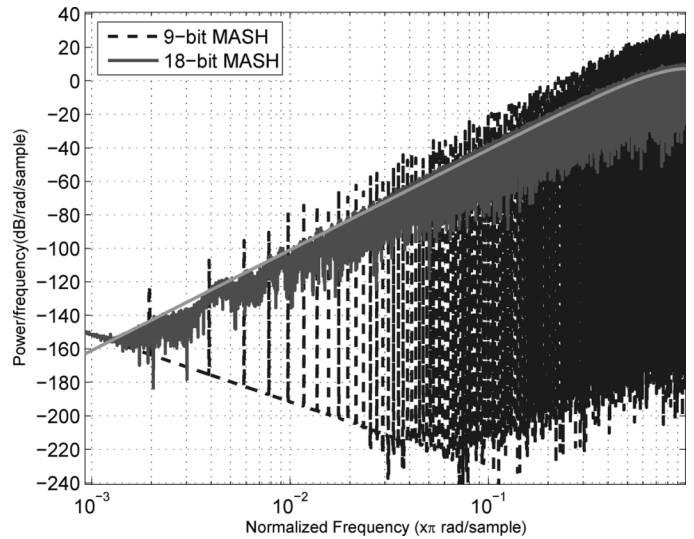


Fig. 9. Output spectra of odd seeded MASH-1–1–1 DDSMs for 9 and 18 bit word lengths with normalized input 0.5. The solid curve represents the shaped white quantization noise.

and have performed extensive simulations on various orders of MASH and EFMs with different seed values in order to obtain guaranteed minimum sequence lengths. They have shown that the modulator produces a guaranteed minimum sequence length when the initial condition of the first stage is constrained to be odd. Some of their empirical results were reported in Table I; we have proven these results for the cases of first, second, and third-order modulators ($l = 1, 2, 3$) [13].

To approach a smooth spectrum, the modulator word length, n_0 , should be increased. Fig. 9 shows clearly that the spectrum of the 18-bit DDSM, which has a length of 2^{19} , is significantly smoother than that of the 9-bit modulator whose quantization noise is distributed over 2^{10} tones. The 18-bit spectrum is comparable to that in the 9-bit dithered case at high frequencies but with superior low-frequency performance.

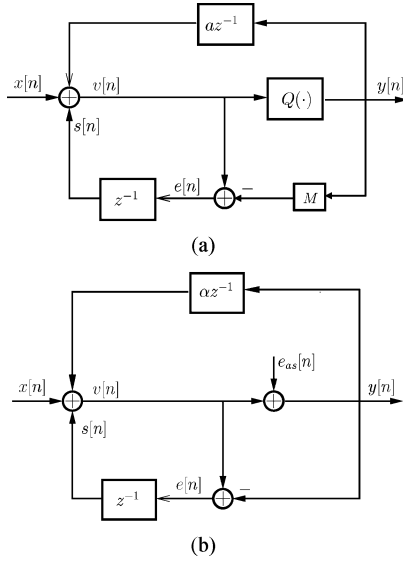


Fig. 10. Modified EFM (a) and (b) its linearized model.

In the rest of this paper, we describe a novel deterministic MASH structure that guarantees a sequence length of approximately 2^{ln_0} for *all* constant inputs and for *all* initial conditions, where l is the order of the MASH. It achieves the spectral shaping of a dithered structure with the same word-length but without the corresponding degradation in the low-frequency performance. For example, we will show that our 9-bit structure outperforms a conventional 9-bit dithered MASH.

III. NEW STRUCTURE

Fig. 10(a) shows our modified architecture for the first-order EFM [16]. This structure differs from the conventional EFM shown in Fig. 4 in that it includes a feedback block az^{-1} , where a is a specially-chosen small integer, to be discussed later. We will show that this modification yields the maximum sequence length for all constant inputs, independently of the initial conditions.

The quantizer input–output relationship is determined by (1), which is the same as the quantizer operation of the conventional structure. In Fig. 10(b), the quantizer is replaced by an additive noise source (e_{as}) to investigate the overall operation of the structure in the z -domain. Note that, for the purpose of the linear analysis, all the signals shown in Fig. 10(b) are normalized to 1. The parameter α is the normalized version of a given by

$$\alpha = \frac{a}{2^{n_0}}. \quad (6)$$

We next derive an expression for the output $y[n]$ in terms of the input $x[n]$ and quantizer error $e_{as}[n]$. In the z -domain

$$Y(z) = V(z) + E_{as}(z) \quad (7)$$

$$E(z) = V(z) - Y(z) \quad (8)$$

and therefore we have

$$E(z) = -E_{as}(z). \quad (9)$$

TABLE II
CORRESPONDING VALUES OF a WITH RESPECT TO THE MODULATOR WORD LENGTH, FOR n_0 IN THE RANGE 5 TO 25 [SEE (14) AND (22)]

n_0	a
5, 7, 13, 17, 19	1
6, 9, 10, 12, 14, 20, 22, 24	3
8, 18, 25	5
11, 21	9
16, 23	15
15	19

At the input summing node, we obtain

$$V(z) = X(z) + \alpha z^{-1}Y(z) + z^{-1}E(z). \quad (10)$$

Substituting (9) into (10), we obtain

$$V(z) = X(z) + \alpha z^{-1}Y(z) - z^{-1}E_{as}(z). \quad (11)$$

Using the value of $V(z)$ from (7) and substituting it into (11), we have

$$Y(z) - E_{as}(z) = X(z) + \alpha z^{-1}Y(z) - z^{-1}E_{as}(z). \quad (12)$$

Rearranging (12), we can write

$$Y(z) = \frac{X(z)}{1 - \alpha z^{-1}} + \frac{1 - z^{-1}}{1 - \alpha z^{-1}}E_{as}(z). \quad (13)$$

The signal and noise transfer functions are $1/(1 - \alpha z^{-1})$ and $(1 - z^{-1})/(1 - \alpha z^{-1})$ respectively. Compared to a conventional first-order digital EFM [see (3)], a pole at $z = \alpha$ is added to both the signal transfer function (STF) and the noise transfer function (NTF). When normalized to 1, if α is sufficiently small, this pole is very close to the origin in the z plane; equivalently, it is a distant pole which does not significantly affect the overall operation of the modulator in the in-band. From a noise-shaping point of view, however, it suppresses the high-frequency noise close to $f_s/2$.

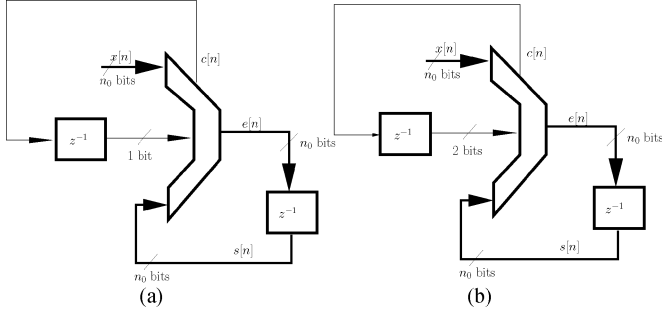
What is special about this architecture compared to the conventional EFM? In Appendix I, we prove that the sequence length of the structure shown in Fig. 10(a) is

$$N = 2^{n_0} - a = M - a \quad (14)$$

for *all* constant inputs and for *all* initial conditions, where n_0 is the accumulator word length and a is determined from Table II. The table can be extended to $n_0 > 25$ by choosing a as the difference between $M = 2^{n_0}$ and the closest prime number less than M .

As an example, for a word-length of 17 ($n_0 = 17$), the structure gives a sequence length of $2^{17} - 1$ for all constant inputs and for all initial conditions; this is approximately equal to the longest possible sequence length 2^{17} . In a conventional structure, the maximum sequence length (2^{17}) is only achievable for *some* combinations of digital inputs and initial conditions [13].

Possible digital implementations when $a = 1$ and $a = 3$ are shown in Fig. 11. In Fig. 11(a), a delayed version of the output is used as the additional feedback signal. In the case of

Fig. 11. Digital implementation of the modified EFM (a) $a = 1$ and (b) $a = 3$.

$a = 1$, this signal is simply applied to the carry-in input of the accumulator. For other values of a determined from Table II, $c[\cdot]$ is used to generate the binary equivalent of a . For example, in Fig. 11(b), the two-bit signal applied to the carry-in input assumes the values 0 or 3. When $y[n] = 1$, a is added to the input and 0 is added to the input when $y[n] = 0$.

Like the conventional MASH structure, we can build a modified MASH (HK-MASH) using this modified first-order modulator.

IV. HIGHER ORDER MODIFIED MASH ARCHITECTURES

Using the structure introduced in the previous section, one can construct higher order MASH DDSMs [16], as shown in Fig. 12. In this figure, each block labeled MEFM1 contains the modified EFM structure shown in Fig. 10(a). Using (13), we can calculate the z -domain input–output relationship. Note that $e[n] = -e_{as}[n]$ in Fig. 10. Therefore

$$Y_1(z) = \frac{X(z)}{1 - \alpha z^{-1}} - \frac{1 - z^{-1}}{1 - \alpha z^{-1}} E_1(z) \quad (15)$$

$$Y_2(z) = \frac{E_1(z)}{1 - \alpha z^{-1}} - \frac{1 - z^{-1}}{1 - \alpha z^{-1}} E_2(z) \quad (16)$$

and, for the l th stage, we obtain

$$Y_l(z) = \frac{E_{(l-1)}(z)}{1 - \alpha z^{-1}} - \frac{1 - z^{-1}}{1 - \alpha z^{-1}} E_l(z). \quad (17)$$

Using the above equations, and looking at Fig. 12, we write

$$Y_M(z) = Y_1(z) + Y_2(z)(1 - z^{-1}) + \dots + Y_{l-1}(z)(1 - z^{-1})^{l-1} \quad (18)$$

$$Y_M(z) = \frac{X(z)}{1 - \alpha z^{-1}} - \frac{(1 - z^{-1})^l}{1 - \alpha z^{-1}} E_l(z) \quad (19)$$

and finally, we have

$$Y(z) = X(z) - (1 - z^{-1})^l E_l(z) \quad (20)$$

or

$$Y(z) = X(z) + (1 - z^{-1})^l E_{asl}(z). \quad (21)$$

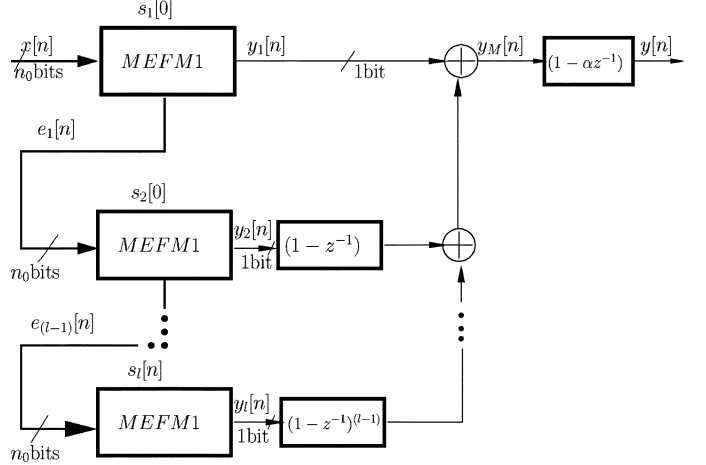


Fig. 12. HK-MASH structure comprising cascaded modified EFMs.

TABLE III
A COMPARISON OF GUARANTEED MINIMUM SEQUENCE LENGTHS OF OUR STRUCTURE AND THE CONVENTIONAL EFM MASH [12] AND [13]

Modulator Order l	[12], [13] odd initial condition on the 1st stage	This work all initial conditions all constant digital inputs
2	$2^{(n_0-1)}$	$(2^{n_0} - a)^2 \approx (2^{2n_0})$
3	$2^{(n_0+1)}$	$(2^{n_0} - a)^3 \approx (2^{3n_0})$
4	$2^{(n_0+1)}$	$(2^{n_0} - a)^4 \approx (2^{4n_0})$
5	$2^{(n_0+2)}$	$(2^{n_0} - a)^5 \approx (2^{5n_0})$

Using this structure, we obtain the desired fraction and the high-order noise-shaping operation of the conventional MASH modulator described in Section II [see (5)].

In Appendix II, we prove that the sequence length of the HK-MASH structure shown in Fig. 12 is determined by the following equation:

$$N = (2^{n_0} - a)^l \quad (22)$$

where the value of a is determined from Table II. For example, in the case of $n_0 = 19$ and $l = 3$, referring to Table III and (22), we have a sequence length of $(2^{19} - 1)^3$ for all constant digital inputs and for all initial conditions. We note that these results are always achieved, without seeding or dithering.

Table III compares the results of this work with the state of the art results in [12] and [13]; a is determined from Table II. Note that adding a stage to the HK-MASH structure increases the sequence length by approximately 2^{n_0} compared to a factor of no more than 4 in the conventional case. Note also that the error introduced by omitting the filter $(1 - \alpha z^{-1})$ is negligible. Therefore, in the following section, we consider the signal $y_M[n]$ rather than $y[n]$.

V. PERFORMANCE COMPARISON OF DIFFERENT MAXIMIZED MASH DDSMS

A. Spectral Investigation

1) *HK-MASH Versus Seeded MASH*: In order to verify the effect of randomization achieved by maximizing the se-

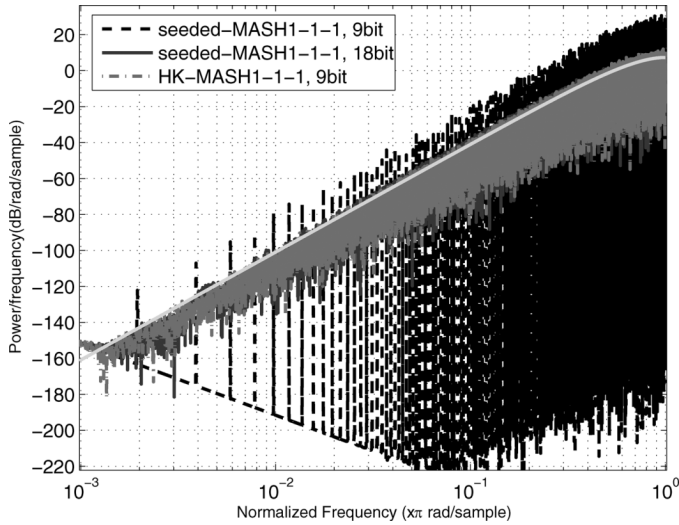


Fig. 13. Output spectra of a 9-bit HK-MASH1-1-1, a 9-bit odd seeded MASH1-1-1 and a 18-bit odd seeded MASH1-1-1 with normalized input 0.5. The solid curve represents the shaped white quantization noise.

quence length for high-order MASH modulators, we present the associated power spectra. The spectral performance of the HK-MASH1-1-1 structure for word-lengths greater than 7-bit has been confirmed through simulations. Word-lengths less than 7 do a poor job of whitening the quantization noise. Fig. 13 shows the output spectra of a 9-bit odd seeded MASH1-1-1 [12], an 18-bit odd seeded MASH1-1-1 [12] and a 9-bit HK-MASH1-1-1, all with normalized input 0.5. Comparing the curves, the 9-bit HK-MASH1-1-1 and the 18-bit odd seeded MASH1-1-1 have almost indistinguishable smooth spectra while the spectrum of the 9-bit odd seeded MASH1-1-1 contains a large number of discrete tones, as expected. Thus, a 9-bit HK-MASH is as effective as an 18-bit seeded MASH.

2) *HK-MASH Versus Dithered MASH*: Fig. 14 shows the spectra of a 9-bit dithered MASH1-1-1 and a 9-bit HK-MASH1-1-1 with unnormalized input 1. A binary pseudorandom dither sequence was generated using a 21-bit linear feedback shift register. This was added to the LSB of the input after being shaped by the filter $(1 - z^{-1})$. Note that both techniques maximize the sequence length effectively; the spectra are almost indistinguishable at high frequencies. However, in the case of the dithered MASH1-1-1, the in-band noise is increased due to the additional first-order noise shaped dither component. As expected, the 9-bit HK-MASH is more effective than this dithered 9-bit MASH at low frequencies.

More sophisticated dithering schemes may be used to shape the low-frequency components but first-order is the recommended limit for LSB dithering at the input [10]. At best, the most effective dithering scheme will also reach the shaped white quantization noise curve shown solid in Fig. 14.

Fig. 15 shows experimentally measured results for a 9-bit odd seeded MASH1-1-1, an 18-bit odd seeded MASH1-1-1 and a 9-bit HK-MASH1-1-1 with a normalized input $X = 0.5$. The modulators were implemented on a Xilinx Spartan II FPGA. The digital outputs in both cases are applied to an Analog Devices Eval-AD5424 DAC board. The spectra were measured using an Agilent EE4402B spectrum analyzer.

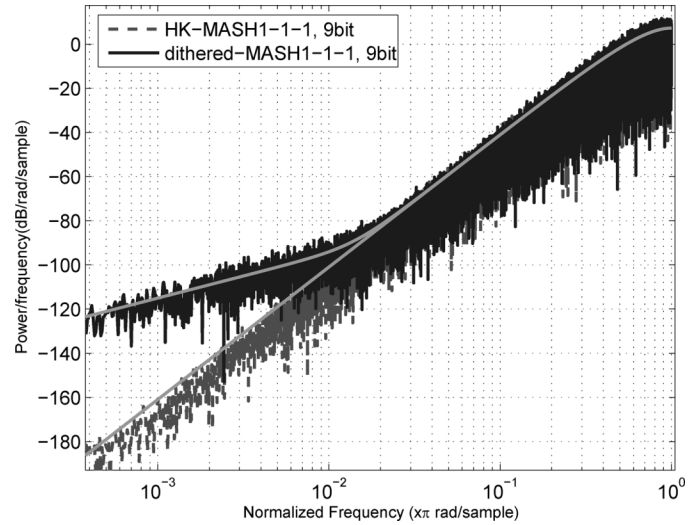


Fig. 14. Spectra of a 9-bit HK-MASH1-1-1 and a 9-bit shaped dithered MASH1-1-1 with input decimal 1. The solid curves represent the shaped white quantization noise including the dither contribution in the case of the dithered MASH DDSM.

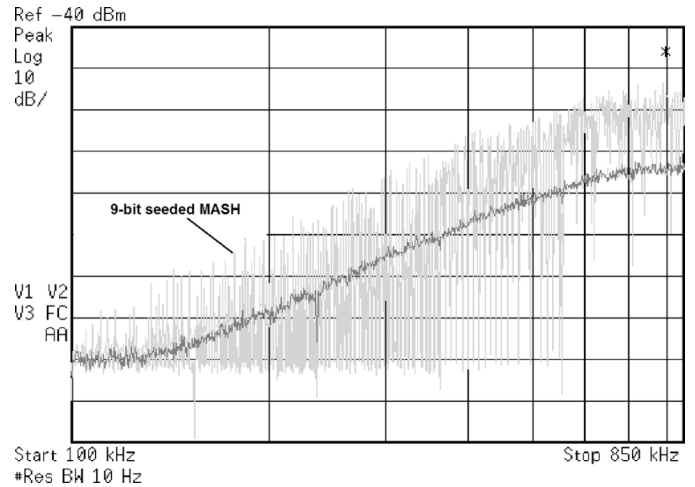


Fig. 15. Measured spectra of a 9-bit HK-MASH1-1-1 (blue), a 9-bit odd seeded MASH1-1-1 (green) and a 18-bit odd seeded MASH1-1-1 (red) with normalized input 0.5.

The measured results confirm qualitatively the theory and simulations. The measured spectrum of the 9-bit MASH in Fig. 15 contains a large number of discrete tones. These result from the short nonwhite quantization noise sequence. As expected, the increase in the sequence length in the 9-bit HK-MASH distributes the quantization noise power over many more frequency components. Consequently, the measured spectrum is much smoother over the entire frequency range, as predicted by the simulations. The 18-bit MASH and 9-bit HK-MASH are almost indistinguishable, as expected.

Fig. 16 shows measured results for a 9-bit first-order shaped LSB dithered MASH1-1-1, an 18-bit first-order shaped LSB-dithered MASH1-1-1 and a 9-bit HKMASH1-1-1 with constant input 1. The spectra are almost smooth and matched over the range shown, except for some small low-frequency spurs on the spectrum of the 9-bit dithered MASH1-1-1. The experiment confirms that the 9-bit HK-MASH is as good as an 18-bit MASH, and better at low frequencies than the dithered

TABLE IV
HARDWARE COMPARISON OF FIRST ORDER SHAPED DITHERED MASH1-1-1 AND HK-MASH1-1-1;
THE NUMBERS GIVEN IN THE 3RD AND 4TH COLUMNS ARE THE TOTAL EQUIVALENT GATE COUNTS

Word-Length	a	Dithered MASH	HK-MASH	$\frac{\text{HK-MASH}}{\text{Dithered MASH}} \%$
9	3	1102	1091	99
11	9	1270	1295	102
13	1	1438	1238	86
15	19	1606	1721	107
17	1	1774	1574	89
19	1	1942	1742	90
24	3	2362	2621	111

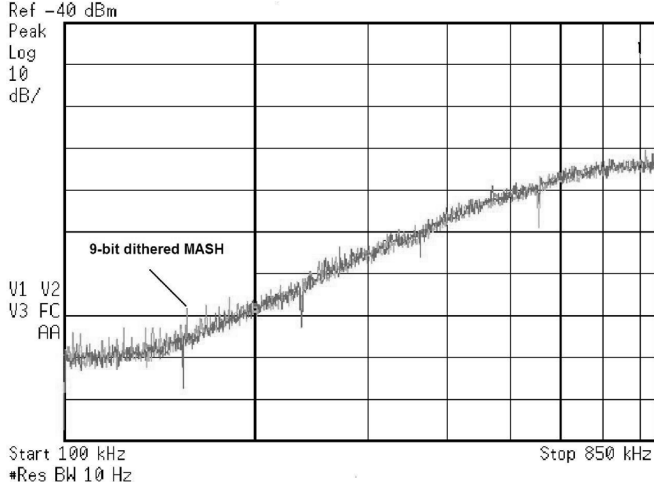


Fig. 16. Spectra of a 9-bit third-order HK-MASH1-1-1 (blue) and a 9-bit shaped dithered MASH1-1-1 (green) and a 18-bit shaped dithered MASH1-1-1 (red) with input 1.

9-bit MASH. The low-frequency sections of the spectra are flattened due to the noise floor of the measuring instrument so we have not been able to confirm experimentally the difference predicted in Fig. 14.

B. Hardware Comparison

A 21-bit linear feedback shift register circuit has been designed and its 1-bit output is applied to the carry-in of the input adder of the second stage of the MASH-1-1-1 DDSM. This ensures a compact realization of the first-order shaped LSB dithering added to the input described in [10]. In this case, there is no need to apply the generated dither sequence to the filter $(1 - z^{-1})$ nor to instantiate an extra adder to add the resulting signed shaped dither sequence to the input of the modulator. For the HK-MASH1-1-1 structure the implementation is the simplest when $a = 1$, as the 1-bit feedback sequence from the output of the block az^{-1} in Fig. 10 is simply applied to the carry-in of the input adder. One approach for the HK-MASH DDSM when $a \neq 1$ is to implement another adder for each stage with the size of $n_0 + 1$ while the summation result of $x[n]$ and $s[n]$ is applied to one input and the 2- to 5-bit feedback sequence from the output of the block az^{-1} is applied to the LSBs of the other input whose MSBs are set to zero.

Hardware synthesis has been performed using the Xilinx ISE 8.2i tool, for the modulators described above with word-lengths

7 to 25. Sample results are reported in Table IV. For word lengths 13, 17, and 19, the HK-MASH has less hardware than the dithered-MASH. The HK-MASH DDSM consumes almost the same amount of hardware as the dithered-MASH for word-lengths 9 and 11. For word-lengths 15 and 24, dithered MASH has 7% and 11% less hardware respectively when compared to 15- and 24-bit HK-MASH DDSMs.

VI. CONCLUSION

In this paper, we have introduced a modified first-order EFM which yields the maximum sequence length for *all* constant inputs and for *all* initial conditions. Based on this structure, we have developed a modified high-order MASH (HK-MASH) DDSM, which yields a dramatically increased sequence length for all dc inputs and for all initial conditions compared to a conventional undithered MASH. The sequence length is $\approx 2^{n_0 l}$, where n_0 is the word-length and l is the number of first-order stages. We have proven this result mathematically.

Simulations and experiments have been performed in order to observe the effect of maximized sequence lengths on the noise-shaping spectra of these modulators. Simulations suggest that for the HK-MASH1-1-1 DDSM, word-lengths as small as 9 achieve smooth spectra resulting from maximum length randomized sequences.

The HK-MASH DDSM provides a deterministic approach that compares favorably with state of the art dithering strategies for maximizing sequence lengths. While consuming almost the same amount of hardware, it has a better low-frequency noise floor.

In this work, we have proven results only for the case of constant inputs. The theoretical component of this research project related to nonconstant inputs is ongoing.

APPENDIX I PROOF OF THE SEQUENCE LENGTH FOR MODIFIED FIRST-ORDER DSM

In this paper, we have introduced a modified EFM structure which gives the maximum sequence length for all constant digital inputs and for all initial conditions. We consider the modified EFM shown in Fig. 10(a). We have that

$$e[n] = v[n] \bmod M \quad (23)$$

where $M = 2^{n_0}$ and n_0 is the accumulator word length. By looking at Fig. 10(a), we write (23) as

$$e[n] = (x[n] + e[n-1] + ay[n-1]) \bmod M. \quad (24)$$

Expanding (24) with its indices, we have

$$e[0] = (x[0] + s[0]) \bmod M \quad (25)$$

where $s[0]$ is the initial condition. Continuing with the other indices, we obtain

$$e[1] = (x[1] + e[0] + ay[0]) \bmod M \quad (26)$$

$$e[2] = (x[2] + e[1] + ay[1]) \bmod M \quad (27)$$

and for the indexes $n-1$ and n we have

$$e[n-1] = (x[n-1] + e[n-2] + ay[n-2]) \bmod M \quad (28)$$

$$e[n] = (x[n] + e[n-1] + ay[n-1]) \bmod M. \quad (29)$$

If we replace the value of each $e[i]$ with its value from the previous equation and recall that

$$(a + b \bmod M) \bmod M = (a + b) \bmod M \quad (30)$$

for positive integers a and b [11], we can write $e[n]$ as

$$e[n] = \left(s[0] + \sum_{k=0}^n x[k] + a \sum_{k=0}^{n-1} y[k] \right) \bmod M \quad (31)$$

which, in the case of a constant dc input X , gives

$$e[n] = \left(s[0] + (n+1)X + a \sum_{k=0}^{n-1} y[k] \right) \bmod M. \quad (32)$$

Before starting the procedure of finding the period of (31), we find a relationship between $x[n]$ and $y[n]$. By looking at Fig. 10(a), we write

$$\begin{aligned} e[n] &= v[n] - My[n] \\ &= ay[n-1] + x[n] + e[n-1] - My[n] \end{aligned} \quad (33)$$

which we can rewrite as

$$My[n] - ay[n-1] = x[n] + e[n-1] - e[n]. \quad (34)$$

Expanding (34) with its indexes yields

$$My[1] - ay[0] = x[1] + e[0] - e[1], \quad (35)$$

$$My[2] - ay[1] = x[2] + e[1] - e[2], \quad (36)$$

$$\vdots \quad (37)$$

$$My[n] - ay[n-1] = x[n] + e[n-1] - e[n]. \quad (38)$$

If we add the left side terms together and do the same for the right side terms of the above equations, we obtain

$$\sum_{k=1}^n My[k] - a \sum_{k=0}^{n-1} y[k] = \sum_{k=1}^n x[k] + e[0] - e[n]. \quad (39)$$

If the system is periodic in the steady state with period N , we have

$$\sum_{k=1}^N My[k] - a \sum_{k=0}^{N-1} y[k] = \sum_{k=1}^N x[k] + e[0] - e[N] \quad (40)$$

where $e[0] = e[N]$, and

$$\sum_{k=1}^N y[k] = \sum_{k=0}^{N-1} y[k]. \quad (41)$$

Therefore, in the steady state, we obtain

$$(M-a) \sum_{k=1}^N y[k] = \sum_{k=1}^N x[k]. \quad (42)$$

Rearranging (42) gives

$$\sum_{k=1}^N y[k] = \frac{1}{M-a} \sum_{k=1}^N x[k]. \quad (43)$$

For a constant dc input X , we have

$$\sum_{k=1}^N y[k] = \frac{NX}{M-a}. \quad (44)$$

Equation (43) plays an essential role in our analysis.

Now we return to find N , the period of the system. In order for (32) to be periodic with period N , we require that

$$e[n] = e[n+N]. \quad (45)$$

Substituting from (32) for $e[n]$ and $e[n+N]$ in (45), we obtain

$$\left(NX + a \sum_{k=1}^N y[k] \right) \bmod M = 0 \quad (46)$$

which is independent of the initial condition, $s[0]$. Using (44) in (46), we obtain

$$\begin{aligned} \left(NX + a \frac{NX}{M-a} \right) \bmod M &= 0 \\ \left(\frac{NX}{M-a} \right) M \bmod M &= 0. \end{aligned} \quad (47)$$

If we choose $M - a$ to be prime and if $0 < X < (M - a)$, the minimum nonzero solution for N such that the above equation is valid

$$N = M - a, \text{ where } N \text{ is prime.} \quad (48)$$

This solution is correct regardless of the input X and the initial condition. The difference between M and the closest prime number to it is a , the values of which, for different modulator word lengths n_0 , are shown in Table II.

APPENDIX II

PROOF OF THE SEQUENCE LENGTH FOR MODIFIED HIGH-ORDER DSM

We extend the results of Appendix I to find the sequence length of the higher order modified MASH modulator shown in Fig. 6. We have proven that, for the first-order modulator, the sequence length for all digital inputs and for any initial condition is N [see (48)], where N is prime. This imposes a restriction on the form of the period of the higher order modulators. If the higher order modulator is periodic with a period N_l , then we require that

$$N_l = KN \quad (49)$$

where N is prime and is determined by (48), and K is an integer.

Modified Second Order Modulator: We use (43) to find the sequence length N_2 of a second-order modulator. Considering Fig. 6, and using (49) and (48), we write

$$\begin{aligned} \sum_{k=1}^{N_2} y_2[k] &= \left(\frac{1}{M-a} \right) \sum_{k=1}^{N_2} e_1[k] \\ \sum_{k=1}^{KN} y_2[k] &= \left(\frac{K}{M-a} \right) \sum_{k=1}^N e_1[k] \\ &= \left(\frac{K}{M-a} \right) \sum_{k=1}^N e_1[k] \\ &= \left(\frac{K}{N} \right) \sum_{k=1}^N e_1[k]. \end{aligned} \quad (50)$$

The only solution for K so that the right side is an integer is N , provided that the greatest common divisor between N and $\sum_{k=1}^N e_1[k]$ is 1. This condition is true; otherwise the solution for K is not unique. With this solution for K , (49) yields

$$N_2 = N^2 \quad (51)$$

where N is prime, and determined by (48).

Modified Third Order and Higher Order Modulators: The solution found in the previous section is used in this section to

find the sequence length of the third stage. The solution for N_3 should be in the form of

$$N_2 = KN^2 \quad (52)$$

where N is prime and K is an integer. Rewriting (43) for the third stage and repeating the same procedure performed for the second stage, we have

$$\sum_{k=1}^{KN^2} y_{3k} = \frac{K}{N} \sum_{k=1}^{N^2} e_{2k}. \quad (53)$$

With the same reasoning for the second stage, the solution for K is N and therefore we have

$$N_3 = N^3 \quad (54)$$

where N is prime and determined by (48). With the same reasoning as for the second and third stage cases, the solution for the l th-order modulator is

$$N_l = N^l \quad (55)$$

where N is prime and determined by (48). Adding an additional stage to this modified structure increases the sequence length by a factor of N , independently of the initial condition and the constant input.

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